Simulation and Validation of a Fixed-Bed Downdraft Wood Gasifier

One-dimensional kinetic model, two-dimensional axisymmetric extension and parametric study,
benchmarked against Jayah et al. (2003)

Course:	Gasification & Reforming Processes — Fixed-Bed Gasifier assignment
Batch:	Batch 09
Reference paper:	T.H. Jayah, L. Aye, R.J. Fuller, D.F. Stewart (2003), *Computer simulation of a downdraft wood gasifier for tea drying*, Biomass and Bioenergy 25(5), 459–469. doi:10.1016/S0961-9534(03)00037-0
Code:	Python (NumPy, SciPy, Matplotlib); repository and `run_simulation.py`
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1 Abstract

The 80 kW_{th} throat-type downdraft gasifier built by the NERD Centre (Sri Lanka) for tea drying and studied by Jayah et al. (2003) was simulated in Python. A throat sub-model (all oxygen consumed by the volatiles, water-gas-shift equilibrium, temperature from an enthalpy balance anchored to the paper's own throat value) feeds a steady one-dimensional kinetic model of the 22 cm conical reduction zone with four reactions (Boudouard, water-gas, methanation, water-gas shift), separate gas and char temperatures, shrinking particles and wall heat loss. A two-dimensional axisymmetric extension resolves radial dispersion of heat and species to give the required contour plots.

Three parameters were calibrated: the wall coefficient, set so that the hearth loses the 12.8 % of input energy Jayah et al. measured, and two char-reactivity factors fitted to Jayah Table 5. The calibration run is matched to a mean absolute error (MAE) of 0.43 %-points, against 0.46 for Jayah's model. Against the nine independent experimental runs (Jayah Table 3) the MAE is 1.42 (CO), 2.01 (H_2), 0.62 (CO_2) and 2.84 %-points (N_2), and the gas heating value is within 0.30 MJ/Nm^3 on average. Run-to-run variations in the data (standard deviations of 0.8–2.4 %-points) are not reproduced, so R^2 about the 1:1 line is low or negative. The predicted axial temperature agrees with the six Jayah Fig. 5 thermocouples to an MAE of 93 K (Jayah's model: 121 K).

A replication of Jayah's parametric study (Jayah Table 7) reproduces the direction of every effect. It does *not* reproduce the 33 cm optimum length: with the measured heat loss imposed, the hearth cools quickly and conversion stops rising after about 11–12 cm. The model therefore points to insulation, not a longer hearth, as the effective improvement.

2 Introduction and problem statement

Sri Lankan tea factories dry leaf with hot air raised by burning fuelwood, mostly rubber wood (*Hevea brasiliensis*). Jayah et al. report that a gasifier-based air heater would cut life-cycle energy cost by about 8 % and wood use by about 12 %. In a downdraft (co-current) gasifier the pyrolysis vapours pass through the hot throat, where most tar is cracked. The gas then reacts with the glowing char in the reduction zone, converting CO_2 and steam into CO and H_2 in endothermic reactions.

The length of that reduction zone is a key design choice. If it is too short, char leaves unreacted; if it is too long, cost and heat loss rise for no benefit. Jayah et al. built a model of the NERD gasifier, calibrated it on their experiments, and recommended extending the zone from 22 to about 33 cm. This assignment reproduces that study, validates the simulation against the paper's measurements, and adds the spatial (radial) information the assignment asks for.

3 Objectives and scope

1. State the feedstock, geometry and operating conditions of the selected paper and use them as simulation inputs.
2. Implement a 1D kinetic model of the reduction zone and a 2D axisymmetric extension, and produce species distributions, species contours and temperature profiles and contours.
3. Calibrate only what the data force us to calibrate, declare it, and validate against data *not* used for calibration (nine runs of Jayah Table 3 and temperatures of Jayah Fig. 5).
4. Repeat the paper's parametric study (chip size, moisture, heat loss, throat angle, air temperature) and compare with its Jayah Table 7.

Operating conditions used. The validation simulations use exactly the conditions reported for each experiment (Jayah Table 3 runs; the Jayah Table 5 calibration run; the Jayah Fig. 5 temperature trial), because validation requires it. The reference case used for profiles and contours is the Jayah Fig. 5 condition (AFR 2.20, 2.5 cm chips, 14.7 % d.b.).

4 Selected paper and related work

Gasifier models range from equilibrium (Gibbs-minimisation) models, which ignore residence time and tend to over-predict CO and H_2 at the low temperatures of the reduction zone, to kinetic and CFD models. Chen (1987) developed a kinetic model of a co-current moving bed with a lumped drying/pyrolysis/combustion zone followed by a kinetic gasification zone. Milligan (1994) replaced the lumped zone with a "flaming pyrolysis" sub-model that gives a more realistic throat temperature and gas composition.

Jayah et al. (2003) combined Milligan's pyrolysis sub-model with Chen's gasification sub-model, modified for the NERD gasifier's diverging throat. They report:

- feedstock analyses (Jayah Tables 1–2);
- nine gas analyses at three chip sizes and several air–fuel ratios (Jayah Table 3);
- a material balance (Jayah Table 4);
- one calibration comparison (Jayah Table 5);
- axial temperatures (Jayah Fig. 5);
- a parametric study summarised in Jayah Table 7.

The paper was chosen because it gives the reactor geometry, the feedstock, and both composition and temperature data along the reduction zone.

5 Reactor and operating conditions

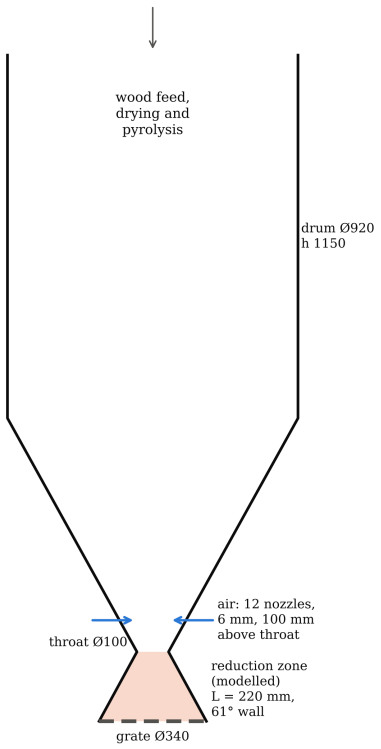


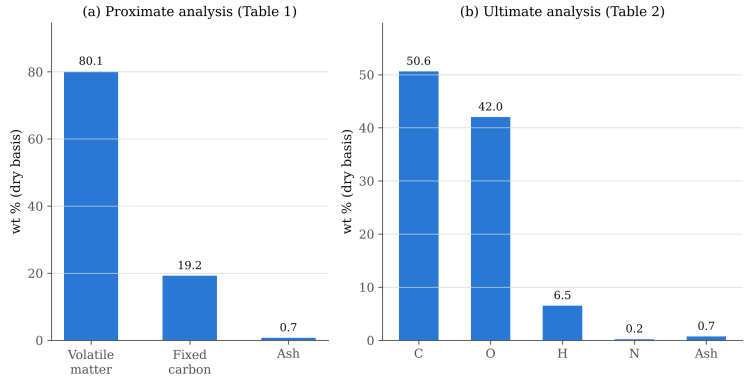
Figure 1: Geometry of the NERD gasifier as modelled (dimensions from Jayah et al. Fig. 3). Only the 220 mm conical reduction zone is resolved spatially; the zones above the throat are represented by the throat sub-model.

Table 1: Geometry and operating data taken from Jayah et al. (2003), with assumptions marked.

Item	Value	Source
Rating	80 kW_{th}	Section 4
Upper drum	0.92 m diameter, 1.15 m long	Section 4
Air inlet	12 nozzles, 6 mm, 100 mm above throat	Section 4
Throat	100 mm diameter, high-alumina castable	Section 4
Reduction zone	220 mm long, wall at 61° from horizontal, 340 mm at the grate	Section 4, Fig. 3
Insulation	none (two steel skins with an air gap)	Section 4
Measured heat loss	12.8 % of input energy	Section 8.4
Chip size	3.3, 4.4, 5.5 cm (runs); 2.5 cm (Jayah Table 5)	Jayah Tables 3, 5
Moisture	12.5–18.5 % d.b.	Jayah Table 3
Air–fuel ratio	1.86–2.37 kg air/kg dry wood	Jayah Table 3
Dry wood feed rate	18.6 kg/h for every run (*assumption*; only Jayah Table 4 gives a feed rate)	Jayah Table 4
Air temperature, pressure	300 K, 1 atm (*assumption*)	—
Bed voidage	0.48 (*assumption*)	—
Char particle density	250 kg/m^3 (*assumption*)	—

With the stoichiometric air requirement of 6.22 kg/kg (calculated from Jayah Table 2), the air–fuel ratios correspond to equivalence ratios of 0.30 (AFR 1.86) to 0.38 (AFR 2.37); the reference case (AFR 2.20) has ER = 0.35.

6 Feedstock and gasification chemistry



(c) Fuel properties: source

Property	Value	Source
HHV (dry)	19.60 MJ/kg	Jayah p. 463 (measured)
LHV (dry)	18.17 MJ/kg	calculated: HHV - 9H*h_fg
Bulk density	314-330 kg/m3	Jayah p. 463 (measured)
Moisture (Table 3 runs)	12.5-18.5 % d.b.	Jayah Table 3
Formula (daf)	CH1.53O0.62N0.0034	calculated from Table 2
Stoich. air	6.22 kg/kg dry	calculated from Table 2
Char density	250 kg/m3	assumption (not in paper)

Figure 2: Rubber wood analyses (Jayah et al. Tables 1–2) and derived properties, with sources.

The dry-ash-free formula is $CH_{1.53}O_{0.62}N_{0.0034}$ and the dry LHV is 18.17 MJ/kg (LHV = HHV − $9\,w_H\,h_{fg}$).

The reduction zone is governed by:

$$C + CO_2 \rightleftharpoons 2\,CO \qquad \Delta H_{298} = +172.5\ \text{kJ/mol} \quad \text{(Boudouard)} \tag{1}$$

$$C + H_2O \rightleftharpoons CO + H_2 \qquad \Delta H_{298} = +131.3\ \text{kJ/mol} \quad \text{(water-gas)} \tag{2}$$

$$C + 2\,H_2 \rightleftharpoons CH_4 \qquad \Delta H_{298} = -74.8\ \text{kJ/mol} \quad \text{(methanation)} \tag{3}$$

$$CO + H_2O \rightleftharpoons CO_2 + H_2 \qquad \Delta H_{298} = -41.2\ \text{kJ/mol} \quad \text{(water-gas shift)} \tag{4}$$

The first two consume char and heat; they set how far conversion proceeds before the bed cools. Reaction enthalpies are evaluated at the local temperature from the species heat capacities (`src/thermo.py`).

7 Mathematical model

7.1 Assumptions

1. Steady state, atmospheric pressure, ideal gas.
2. Above the throat, all air oxygen is consumed by the volatiles. Char (the fixed carbon, 19.2 % of dry wood) enters the reduction zone unreacted, as pure carbon.
3. CH_4 is imposed at 1.2 % of the dry gas leaving the throat, as Jayah et al. did (they set predicted methane equal to measured). CH_4 is therefore not a prediction.
4. Gas and char have separate temperatures (T_g, T_s). The char enters 50 K below the gas (*assumption*).
5. Particles are spheres that shrink as they react; bed voidage is constant.
6. Wall heat loss is $U\,P(z)\,(T_g - T_\infty)$, with P the local wall perimeter.

7.2 Throat sub-model

The volatile C, H and O, the moisture and the air are distributed among CO, CO_2, H_2, H_2O, CH_4 and N_2 by element balances plus water-gas-shift equilibrium. An enthalpy balance on the drying/pyrolysis/flaming zone,

$$\dot m_{\text{wood}}\,\Delta h_{f,\text{wood}} + \dot n_{\text{H}_2\text{O(l)}}\,\Delta h_{f,\text{H}_2\text{O(l)}} + \dot H_{\text{air}}(T_{\text{air}}) = \sum_i \dot n_i \big[\Delta h_{f,i} + h_i(T_{\text{ad}})\big] + \dot n_{\text{C}}\,h_{\text{C}}(T_{\text{ad}} - 50\,\text{K}), \quad (5)$$

gives the mixed-mean adiabatic temperature T_{ad}. The enthalpy of formation of wood is back-calculated from its HHV.

For the Jayah Fig. 5 condition $T_{\text{ad}} = 1345$ K. This is about 260 K below the ≈1604 K at which Jayah's own model curve starts (Jayah Fig. 5, digitised). A fully mixed balance cannot represent the hot near-stoichiometric flame zones at the nozzles. We therefore keep the paper's absolute level and take only the *change* with operating conditions from the balance:

$$T_{g,0} = 1604\,\text{K} + \big[T_{\text{ad}}(\text{AFR},\,\text{MC},\,T_{\text{air}}) - T_{\text{ad}}(2.20,\,14.7\,\%,\,300\,\text{K})\big]. \quad (6)$$

Across the nine runs this gives throat temperatures of 1422–1700 K: higher for more air, lower for wetter wood.

7.3 One-dimensional reduction-zone model

With $A(z) = \pi R(z)^2$, $R(z) = R_{\text{th}} + z\cot 61°$, species molar flows F_i, char flow F_C, and volumetric rates r_j (mol m^{-3} s^{-1}):

$$\frac{\mathrm{d}F_i}{\mathrm{d}z} = A\sum_j \nu_{ij} r_j, \qquad \frac{\mathrm{d}F_C}{\mathrm{d}z} = -A\,(r_1 + r_2 + r_3), \quad (7)$$

$$\Big(\sum_i F_i c_{p,i}\Big)\frac{\mathrm{d}T_g}{\mathrm{d}z} = h_{gs} a_p A\,(T_s - T_g) - A\,r_4\,\Delta H_4(T_g) - U\,P\,(T_g - T_\infty), \quad (8)$$

$$\dot m_C c_{p,C}\frac{\mathrm{d}T_s}{\mathrm{d}z} = -h_{gs} a_p A\,(T_s - T_g) - A\sum_{j=1}^{3} r_j\,\Delta H_j(T_s), \quad (9)$$

$$\frac{\mathrm{d}d_p}{\mathrm{d}z} = -\frac{2M_C}{\rho_{\text{char}} u_s}\,\frac{r_1 + r_2 + r_3}{a_p}, \qquad a_p = \frac{6(1-\varepsilon)}{d_p}, \qquad u_s = \frac{\dot m_C}{\rho_{\text{char}}(1-\varepsilon)A}. \quad (10)$$

The interphase coefficient h_{gs} combines Ranz–Marshall convection, $Nu = 2 + 0.6\,Re_p^{1/2} Pr^{1/3}$, with a radiative term $4\sigma e\,T^3$.

7.4 Kinetics

The heterogeneous rates combine a surface reaction in series with external film mass transfer, driven by the distance from equilibrium:

$$r_j = a_p \frac{p_k - p_k^{\mathrm{eq}}}{\dfrac{RT_s}{\eta_j A_j e^{-E_j/RT_s}} + \dfrac{RT_g}{k_{m,k}}}, \qquad r_4 = \varepsilon\, k_4 \Big(C_{\mathrm{CO}} C_{\mathrm{H_2O}} - \frac{C_{\mathrm{CO_2}} C_{\mathrm{H_2}}}{K_4} \Big), \tag{11}$$

where k_m comes from $Sh = 2 + 1.1\, Re^{0.6} Sc^{1/3}$ and $k_4 = 2.78 \times 10^{-3} \exp(-1510/T_g)\ \mathrm{m^3\,mol^{-1}\,s^{-1}}$.

The effectiveness factors η_j lump intraparticle diffusion and the reactivity of this char. They are the calibrated quantities (Section 7.6); Table 9 lists all values.

The water-gas-shift expression $2.78 \exp(-1510/T)$ is quoted in the literature without consistent concentration units. We read it as $\mathrm{m^3\,kmol^{-1}\,s^{-1}}$, which gives a kinetically slow homogeneous shift. Taking it as $\mathrm{m^3\,mol^{-1}\,s^{-1}}$ instead makes the shift effectively instantaneous and changes the calibration-run gas to CO 9.0, H_2 22.8, CO_2 18.0 %, far from the measurement (19.1, 15.5, 11.4 %). This choice is the most influential assumption in the model and is listed as a limitation.

7.5 Two-dimensional axisymmetric model

The cone is mapped to $\eta = r/R(z) \in [0, 1]$ and split into 12 annular finite-volume cells that follow the wall divergence. Each cell solves the same equations as the 1D model with its own cross-section; only the outermost cell loses heat through the wall. Neighbouring cells exchange species and heat by radial dispersion:

$$\left.\frac{\mathrm{d}F_{i,k}}{\mathrm{d}z}\right|_{\mathrm{disp}} = \Big[2\pi r D_r \frac{\partial C_i}{\partial r} \Big]_{k-1/2}^{k+1/2}, \qquad \Big(\sum F c_p \Big)_k \left.\frac{\mathrm{d}T_{g,k}}{\mathrm{d}z}\right|_{\mathrm{disp}} = \Big[2\pi r k_r \frac{\partial T_g}{\partial r} \Big]_{k-1/2}^{k+1/2}, \tag{12}$$

with packed-bed dispersion coefficients

$$D_r = \varepsilon D_m + \frac{u\, d_p}{Pe_m}, \qquad Pe_m = 10, \tag{13}$$

$$k_r = \varepsilon k_g + \frac{4\sigma e}{2 - e}\, d_p T^3 + \frac{\rho c_p\, u\, d_p}{Pe_h}, \qquad Pe_h = 8. \tag{14}$$

Boundary conditions are symmetry at the axis and an impermeable wall that loses heat at the outer cell. The inlet is radially uniform.

The system (12×10 states) is integrated as one stiff ODE system in z (method of lines). A test confirms that with very strong radial mixing the 2D model collapses onto the 1D model. The model resolves radial heat and mass dispersion only: it has no velocity profile, no wall channelling, no nozzle jets, and is not CFD.

7.6 Calibration

Three parameters were calibrated (`src/calibration.py`); none of the validation data were used.

1. The wall coefficient $U = 114.4\ \mathrm{W\,m^{-2}\,K^{-1}}$ makes the 22 cm hearth lose 12.8 % of the feed's HHV input, the loss Jayah et al. measured. Jayah et al. themselves specify heat loss as a percentage of input energy in their gasification-zone model (their Fig. 9). The value is large for a physical wall coefficient. It should be read as an effective coefficient that assigns all of the measured loss to the uninsulated hearth.

2. The effectiveness factors $\eta_{\mathrm{B}} = 0.474$ (Boudouard) and $\eta_{\mathrm{WG}} = 0.142$ (water-gas) minimise the squared error in CO, H_2 and CO_2 for Jayah Table 5. Jayah Table 5 is the 2.5 cm calibration run, which the paper identifies as Run 7, so Run 7's moisture (14.7 %) and AFR (1.86) are used. Jayah Table 3 lists Run 7 with 5.5 cm chips, an inconsistency in the paper.

7.7 Numerical method

The 1D equations are integrated with SciPy's implicit Radau method (relative tolerance 10^{-5}, absolute 10^{-8}). The 2D system uses Radau with the block-tridiagonal Jacobian sparsity (111 output stations). The complete pipeline runs in about 29 s.

8 Verification

- **Element balances:** C, H, O and N close to machine precision ($< 10^{-12}$ %), because the stoichiometry conserves them exactly.
- **Energy balance:** with total (formation + sensible) enthalpy flows, $\dot{H}(0) - \dot{H}(L) - \dot{Q}_{\text{wall}}$ is -0.07 % of the feed HHV. The residual comes from evaluating heterogeneous reaction heats at T_s.
- **Automated tests (20):**
  - feedstock arithmetic, thermodynamics and reaction rates;
  - conservation;
  - the throat temperature rising with AFR and air pre-heat and falling with moisture;
  - all validation statistics recomputed independently;
  - the 2D→1D limit.

  Run with `python -m pytest tests`.

9 Results: reference case

The reference case is the Jayah Fig. 5 condition: AFR 2.20 (ER 0.35), 2.5 cm chips, 14.7 % d.b. moisture, and the 22 cm hearth.

Table 2: Reference-case results at the hearth exit ($z = 22$ cm), 1D model.

Throat gas temperature $T_{g,0}$ (mixed adiabatic value)	1604 K (1345 K)
Exit gas temperature	838 K
Dry gas: CO / H_2 / CO_2 / CH_4 / N_2	21.30 / 13.49 / 9.03 / 1.05 / 55.13 %
Gas LHV	4.52 MJ/Nm^3
Dry gas flow (at 18.6 kg/h dry wood)	45.4 Nm^3/h
Conversion (cold-gas) efficiency, $\text{LHV}_{\text{gas}}/\text{HHV}_{\text{wood}}$	56.3 %
Char conversion	50.3 %
Exit particle diameter (from 2.5 cm)	1.98 cm
Wall heat loss	14.0 kW (13.8 % of 101.3 kW input)

9.1 Axial species distribution

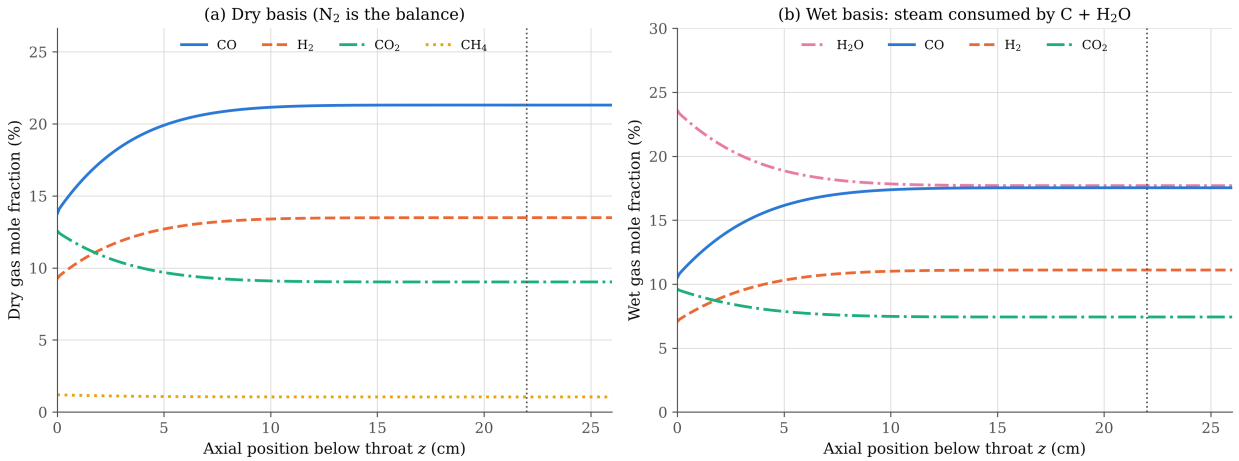


Figure 3: Axial species profiles in the reduction zone (reference case): (a) dry basis, (b) wet basis including steam.

CO rises from 13.74 to 21.30 % and H_2 from 9.24 to 13.49 %, while CO_2 falls from 12.59 to 9.03 % and steam from 23.6 to 17.7 % (wet). Almost all of the change happens in the first 8–10 cm. Beyond that the bed is too cold (below about 1150 K) for the char reactions to proceed, and the composition freezes.

9.2 Char conversion

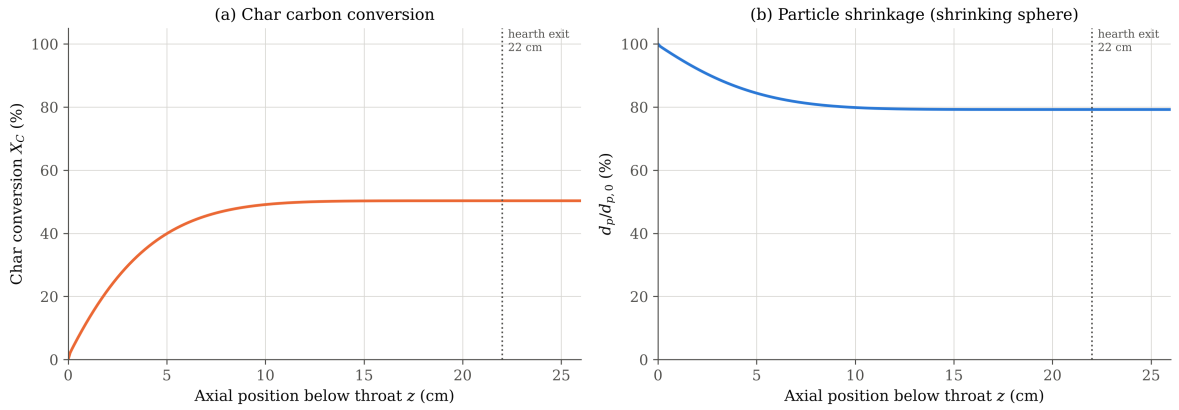


Figure 4: Char conversion and particle shrinkage along the bed (reference case).

Half of the final conversion is reached by $z = 2.4$ cm, and conversion stops at 50.3 %. A quick check against Jayah's Table 4: 0.7–2.1 kg/h of char leaves the grate, out of about 3.6 kg/h of fixed carbon, i.e. 41–80 % conversion. The reference case lies at the lower end of that range. The calibration run (AFR 1.86) converts only 17.0 %, which is low (Section 12).

9.3 Temperature

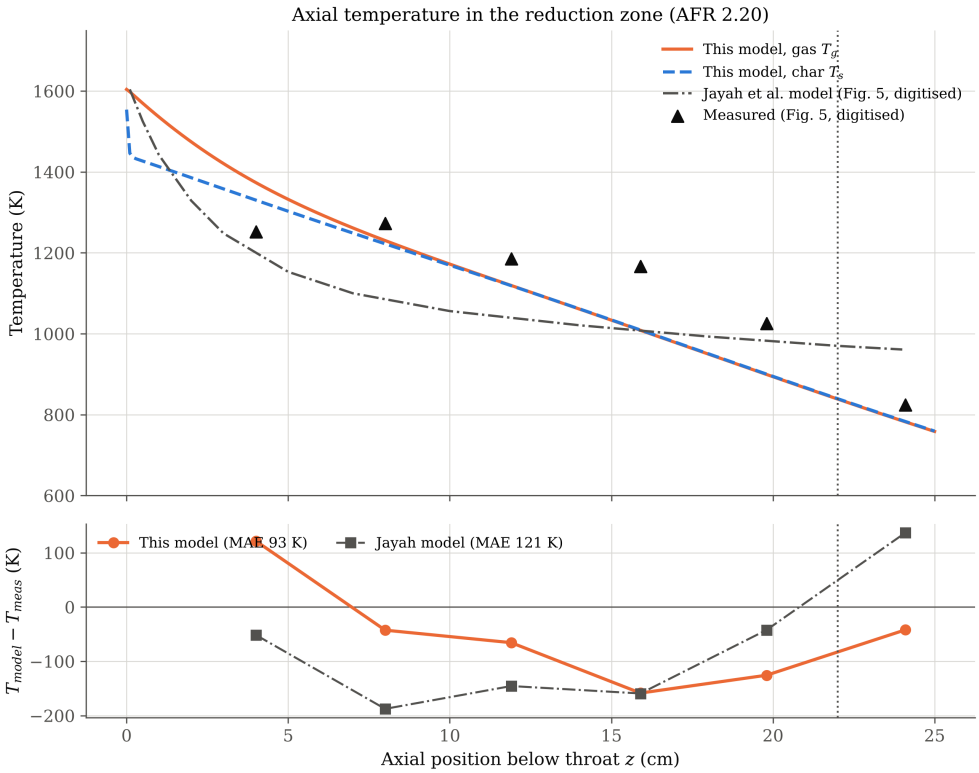


Figure 5: Gas and char temperatures against the six measured points and Jayah's model curve, both digitised from Jayah Fig. 5; the lower panel shows the errors.

The gas enters at 1604 K and the char at 1554 K. The char, with its small heat capacity flow, is cooled immediately by its own endothermic reactions, opening a gas–char gap of up to 155 K. The two phases equilibrate (gap below 10 K) by $z = 7.6$ cm.

The gas cools to 1332 K at 5 cm, 1172 K at 10 cm and 838 K at the exit. Of the 19.9 kW of sensible heat the gas and char lose across the hearth, 71 % leaves through the wall (14.0 kW) and 29 % is absorbed by the endothermic reactions. Section 10 compares these temperatures with the measurements.

9.4 Two-dimensional species and temperature fields

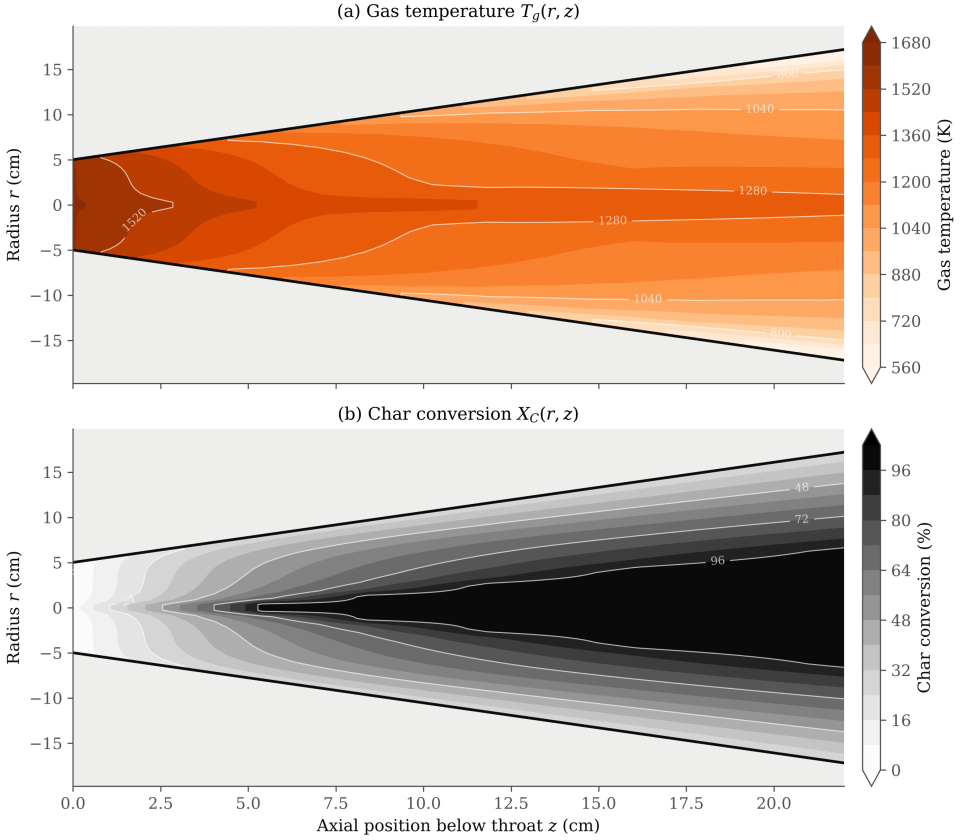


Figure 6: 2D model: gas temperature and char conversion in the conical hearth (mirrored about the axis).

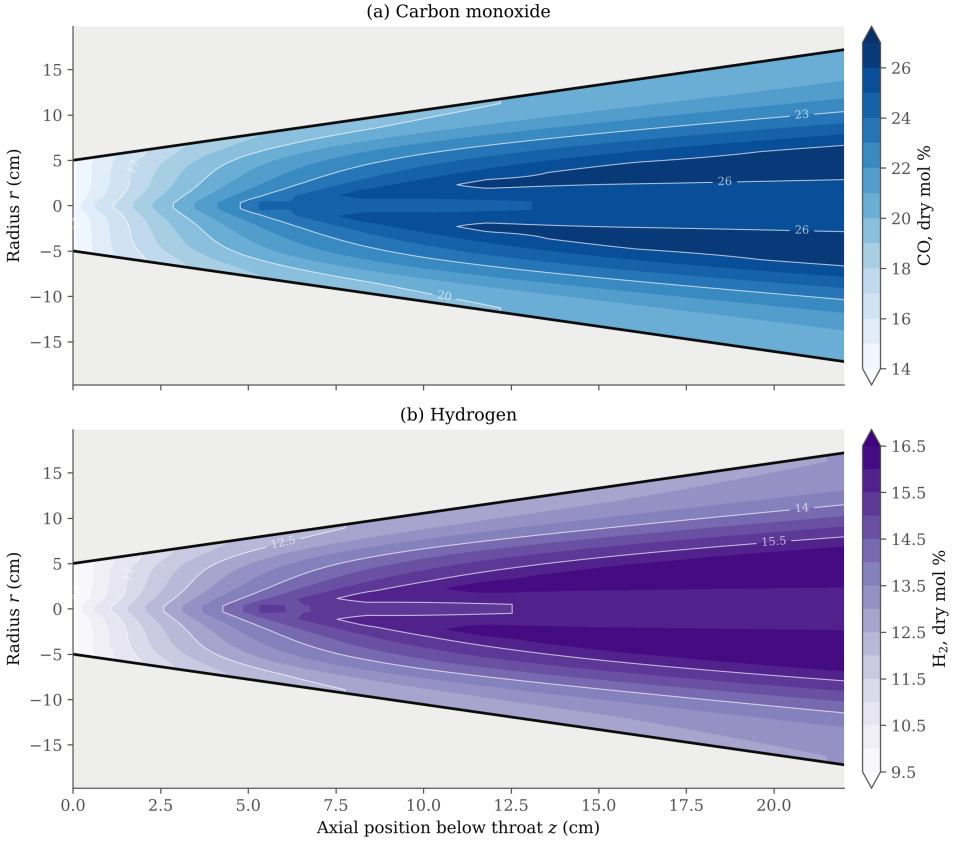


Figure 7: 2D model: CO and H_2 (dry basis).

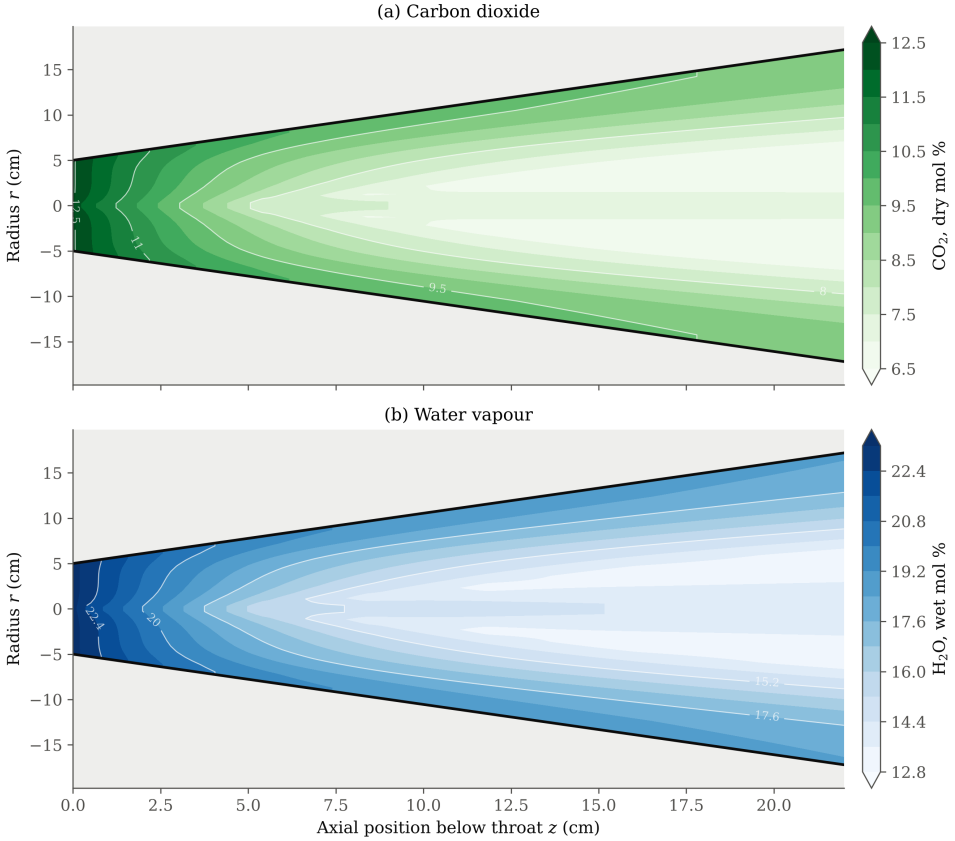


Figure 8: 2D model: CO_2 (dry) and H_2O (wet).

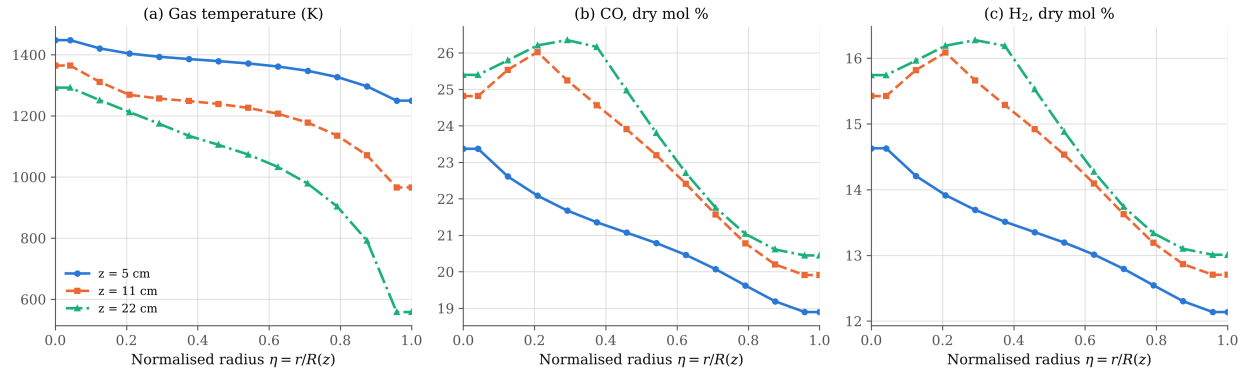


Figure 9: Radial profiles from the 2D model at $z = 5$, 11 and 22 cm.

Because heat leaves only through the wall, a cold boundary layer forms there. At 5 cm the axis is at 1448 K and the wall cell at 1250 K; at the exit the values are 1292 K and 558 K. Gasification consequently proceeds much further in the core than near the wall.

At the exit, CO is 25.4 % on the axis but 20.4 % at the wall, and H_2 is 15.7 against 13.0 %. Char on the axis is 89.5 % converted by 5 cm and 99.8 % by 11 cm, but only 29.9 % near the wall. Once the core char is exhausted, the maximum CO and H_2 move slightly off the axis (Fig. 9), where char is still reacting.

The flow-weighted exit gas of the 2D model (CO 22.76, H_2 14.29, CO_2 8.34, N_2 53.59 %) is richer than the 1D result. The 2D wall loss is lower (10.5 against 14.0 kW) because the wall-side gas is colder. U was calibrated with the 1D (mean-temperature) model, so the 2D results show the *radial structure* of the hearth rather than a separately calibrated prediction.

10 Validation against Jayah et al. (2003)

10.1 Calibration run (Jayah Table 5)

Table 3: Jayah Table 5 run (2.5 cm chips, Run 7 conditions). These data were *used for calibration*, so the agreement is not independent evidence. Errors are in %-points.

Species (dry, %)	Measured	Jayah model	This model	Error, this model	Error, Jayah
CO	19.1	18.3	18.70	-0.40	-0.80
H_2	15.5	16.4	15.65	+0.15	+0.90
CO_2	11.4	11.1	10.72	-0.68	-0.30
CH_4[a]	1.1	1.1	1.14	+0.04	+0.00
N_2	52.9	53.2	53.78	+0.88	+0.30
MAE / RMSE				0.43 / 0.53	0.46 / 0.57

[a]CH_4 imposed at the throat in both models.

10.2 Nine experimental runs (Jayah Table 3): independent validation

Table 4: Measured / simulated dry gas composition (%) for the nine runs, with the simulated throat temperature. Feed rate assumed to be 18.6 kg/h.

Run	Chip (cm)	MC (% d.b.)	AFR	$T_{g,0}$ (K)	CO	H_2	CO_2	N_2
1	3.3	18.5	2.03	1487	19.6 / 18.3	17.2 / 14.1	9.9 / 10.8	51.9 / 55.7
2	3.3	16.0	2.2	1593	20.2 / 20.2	18.3 / 13.1	9.7 / 9.6	50.7 / 56.0
3	3.3	14.7	2.37	1686	19.4 / 21.7	17.2 / 12.6	9.7 / 8.7	52.6 / 55.9
4	4.4	16.0	1.96	1468	18.4 / 18.0	17.0 / 14.3	10.6 / 10.9	52.7 / 55.6
5	4.4	15.2	2.12	1560	19.7 / 19.2	13.2 / 13.1	10.8 / 10.1	55.0 / 56.5
6	4.4	14.0	2.29	1654	18.9 / 20.6	12.5 / 12.3	8.5 / 9.3	59.1 / 56.7
7	5.5	14.7	1.86	1422	19.1 / 17.6	15.5 / 15.1	11.4 / 11.3	52.9 / 54.9
8	5.5	13.8	2.04	1529	22.1 / 18.7	12.7 / 13.3	10.5 / 10.4	53.4 / 56.5
9	5.5	12.5	2.36	1700	19.1 / 20.8	13.0 / 11.8	10.7 / 9.1	56.0 / 57.3

Table 5: Error statistics over the nine runs (%-points; LHV in MJ/Nm^3). R^2 is about the 1:1 line; "Std (meas.)" is the run-to-run spread of the measurements.

	MAE	RMSE	Bias	Std (meas.)	R^2	Within ±10 %	MAE excl. Run 7
CO	1.42	1.74	-0.15	1.00	-1.99	7/9	1.41
H_2	2.01	2.73	-1.87	2.19	-0.55	5/9	2.21
CO_2	0.62	0.78	-0.19	0.80	0.04	7/9	0.68
N_2	2.84	3.07	+2.31	2.39	-0.66	8/9	2.95
CH_4[a]	0.15	0.16	-0.13	0.10	-1.47	3/9	0.16
LHV (MJ/Nm^3)	0.30	0.36	-0.27	0.24	-1.21	7/9	0.32

[a]CH_4 is imposed, so its statistics do not test the model.

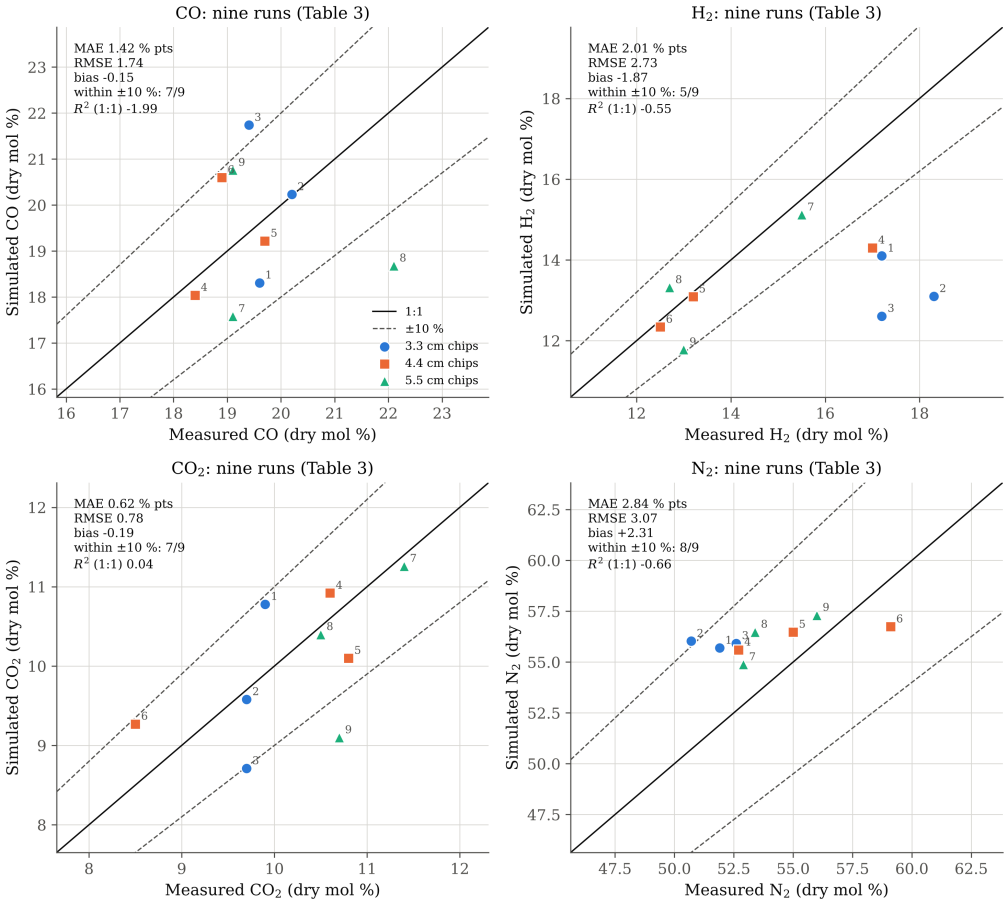


Figure 10: Parity plots for the nine runs; points are labelled by run number. All statistics are computed in `src/validation.py`.

Interpretation. The mean levels are reproduced well. CO has almost no bias (-0.15 %-pts) and CO_2 is within 0.62 %-pts on average, but two systematic problems remain:

- **H_2 is too low** (bias -1.87 %-pts), mainly in Runs 1–4, where the measured H_2 is 17–18 %.
- **N_2 is too high** (bias +2.31), i.e. the model produces less gas per unit of air than the plant did.

The negative R^2 values do not mean the errors are large. The measured run-to-run spread is only 0.8–2.4 %-points, comparable with the model error, and the model does not capture which runs are richer. In the model, a higher AFR raises the throat temperature (Eq. 6) and with it CO. The data show no such trend: CO stays at 18.4–20.2 % for all AFRs, apart from Run 8 at 22.1 %.

Excluding Run 7, whose composition was the calibration target, changes the MAE only slightly (last column).

10.3 Heating value and gas flow

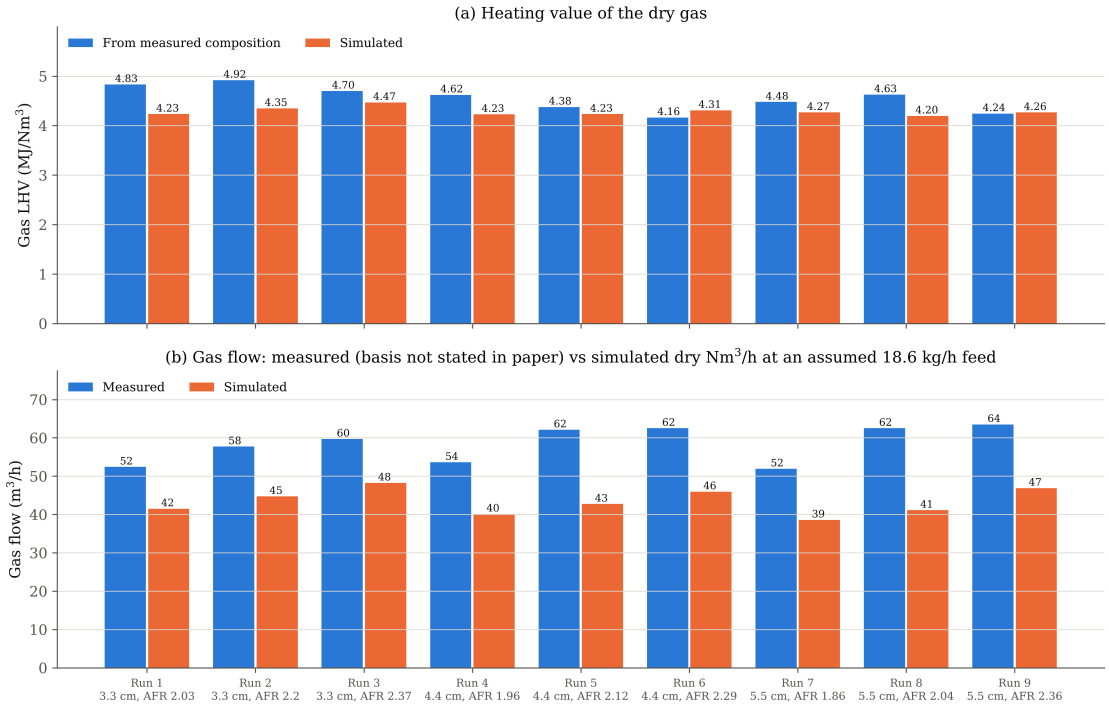


Figure 11: Gas LHV and gas flow per run.

Table 6: LHV, gas flow and conversion efficiency per run. “Implied feed” is the dry feed that would make the simulated flow equal to the measured one.

Run	LHV (MJ/Nm^3) meas.	sim.	Flow meas. (m^3/h)	Flow sim. (Nm^3/h, dry)	Implied feed (kg/h)	Efficiency sim. (%)
1	4.83	4.23	52.4	41.5	23.5	48.2
2	4.92	4.35	57.7	44.7	24.0	53.3
3	4.70	4.47	59.7	48.2	23.0	59.1
4	4.62	4.23	53.6	40.1	24.8	46.5
5	4.38	4.23	62.1	42.7	27.0	49.6
6	4.16	4.31	62.5	45.9	25.3	54.3
7	4.48	4.27	51.9	38.6	25.0	45.2
8	4.63	4.20	62.5	41.1	28.3	47.3
9	4.24	4.26	63.5	46.9	25.2	54.8

The LHV is predicted to within 0.30 MJ/Nm^3 on average (bias -0.27). The simulated dry flow is only 66–81 % of the measured flow, but the comparison is weak. The paper does not state the basis of the measured flow (temperature, wet or dry), and it gives no feed rate for the individual runs. Because gas flow scales with feed, the shortfall corresponds to a dry feed of 23.0–28.3 kg/h instead of the assumed 18.6 kg/h. Gas composition and LHV, being intensive, are almost unaffected by this assumption (Section 11.2).

10.4 Axial temperature (Jayah Fig. 5)

Table 7: Gas temperature along the axis, AFR 2.20 (measured points and Jayah's model digitised from Jayah Fig. 5, ±15 K reading uncertainty).

z (cm)	Measured (K)	This model (K)	Error (K)	Jayah model (K)	Error (K)
4.0	1252	1374	+122	1200	-52
8.0	1273	1230	-43	1085	-188
11.9	1185	1119	-66	1039	-146
15.9	1167	1008	-158	1008	-159
19.8	1025	899	-126	983	-42
24.1	824	782	-42	961	+137
MAE / RMSE			93 / 103		121 / 132

Jayah Fig. 5 shows six measured points. Neither model was fitted to them. The present model is 93 K from the data on average (bias -52 K), better than Jayah's model at 121 K. The largest errors are an over-prediction near the throat (4 cm) and an under-prediction at 16–20 cm, where the measured profile stays flat before dropping. Thermocouple readings in a reacting bed are themselves uncertain: they exchange radiation with the cooler wall and can be fouled by soot.

11 Parametric study and engineering interpretation

11.1 Replication of Jayah et al. Table 7

The base case is 3 cm chips, 15 % d.b., AFR 2.20, air at 300 K, a 61° throat and $U = 114.4\ \mathrm{W\,m^{-2}\,K^{-1}}$; one parameter is varied at a time. For the heat-loss cases, U is re-solved so that the hearth loses 5, 10 or 15 % of the input. The optimum length follows Jayah's definition: the length at which efficiency reaches 99.5 % of its ultimate value.

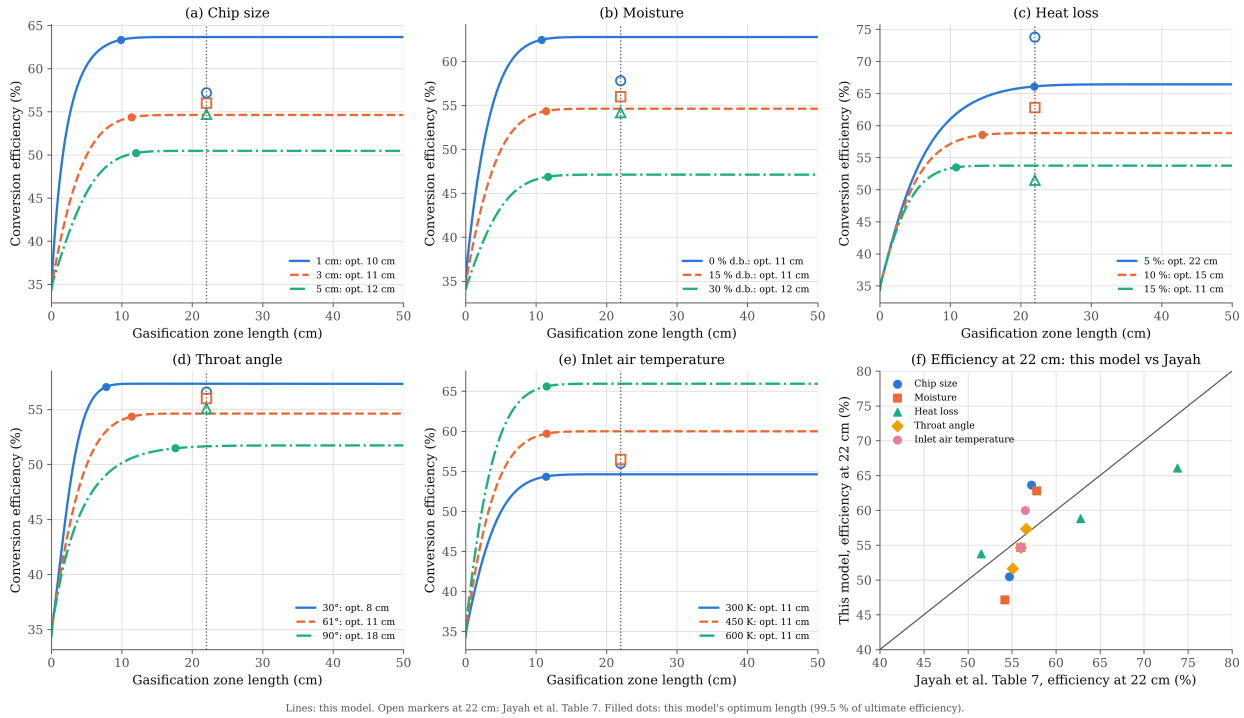


Figure 12: Conversion efficiency vs. gasification-zone length (lines, this model) with Jayah's values at 22 cm (open markers). Panel (f) compares the 22 cm efficiencies directly.

Table 8: Comparison with Jayah et al. Table 7 (efficiency in %, length in cm). The model's 61° case is compared with Jayah's 60° case; "–" means not reported.

Parameter	Value	Efficiency at 22 cm		Optimum length		Efficiency at optimum		Hearth loss
		Model	Jayah	Model	Jayah	Model	Jayah	(% input)
Chip size	1 cm	63.6	57.2	10	24	63.3	57.3	12.8
	3 cm	54.6	56.0	11	32	54.4	56.3	14.0
	5 cm	50.5	54.7	12	33	50.2	55.2	14.6
Moisture	0 % d.b.	62.8	57.8	11	31	62.5	58.1	14.2
	15 % d.b.	54.6	56.0	11	32	54.4	56.3	14.0
	30 % d.b.	47.1	54.2	12	35	46.9	54.5	13.8
Heat loss	5 %	66.1	73.8	22	29	66.1	74.2	5.0
	10 %	58.8	62.8	14	28	58.5	63.2	10.0
	15 %	53.7	51.5	11	25	53.5	51.7	15.0
Throat angle	30°	57.3	56.6	8	28	57.0	57.4	20.9
	61°	54.6	56.0	11	24	54.4	56.2	14.0
	90°	51.6	55.1	18	19	51.5	55.1	7.8
Air temperature	300 K	54.6	56.0	11	–	54.4	–	14.0
	450 K	60.0	56.5	12	–	59.7	–	14.2
	600 K	65.9	–	12	–	65.6	–	14.3

What agrees. The base-case efficiency at 22 cm (54.6 %) is close to Jayah's 56.0 %. Every trend has the right direction:

- smaller chips, drier wood, lower heat loss, a smaller throat angle and hotter air all raise efficiency;
- heat loss is one of the two strongest levers (66.1 % at 5 % loss against 53.7 % at 15 %; Jayah: 73.8 to 51.5 %).

What differs.

- **Moisture, chip size and air temperature act more strongly in this model.** 0–30 % moisture gives 62.8–47.1 % (Jayah 57.8–54.2 %). Pre-heating air to 450 K adds 5 %-points (Jayah: 0.5). These effects act through the throat temperature (Eq. 6) and the strongly temperature-dependent char kinetics, and the size of the response depends on the assumed anchoring.
- **The optimum length is much shorter: 11 cm instead of 32–33 cm.** With 12.8 % of the input lost through an uninsulated hearth, the bed has cooled to 1129 K by 12 cm, where the char reactions (activation energies of about 240 kJ/mol) have effectively stopped. Lengthening the hearth then only adds wall area. Only when the loss is cut to 5 % does the optimum move out to the present 22 cm (22 cm), with efficiency rising to 66.1 %.

Design implication. This model does not support lengthening the hearth to 33 cm on its own. It agrees with Jayah et al. that reducing heat loss (insulating the reactor) is the most effective change. With better insulation, the existing 22 cm hearth is close to the optimum length.

11.2 Air–fuel ratio and feed rate

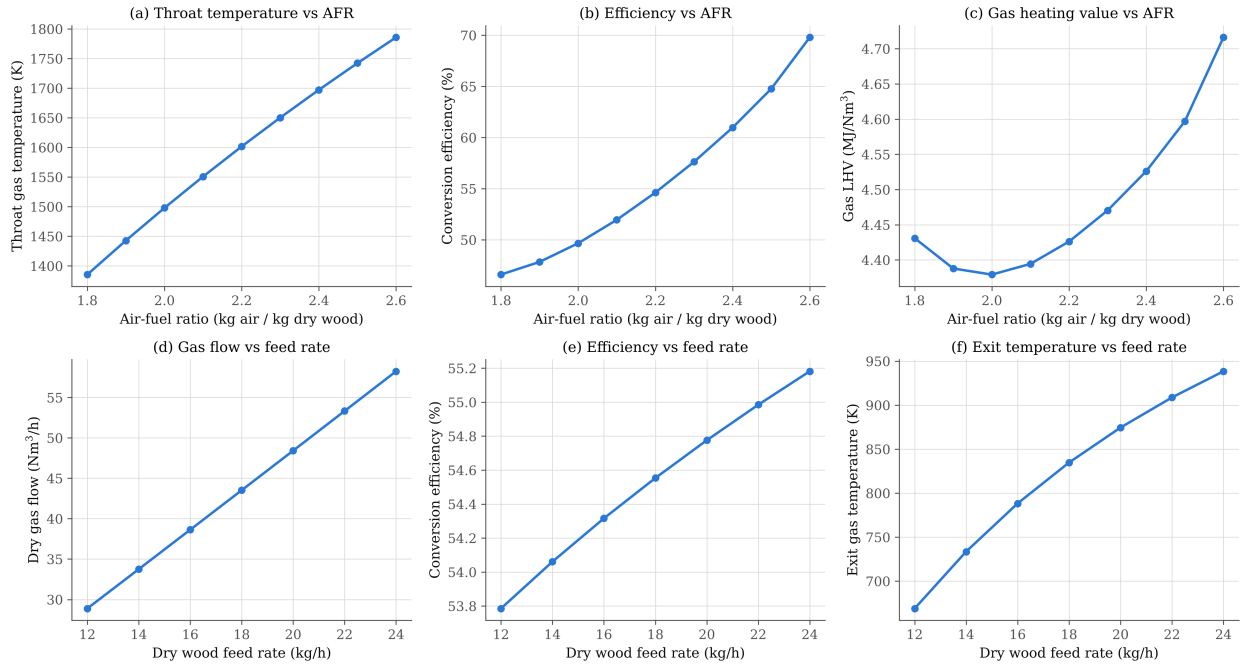


Figure 13: Effect of air–fuel ratio and dry feed rate at 22 cm (3 cm chips, 15 % d.b.).

- **Air–fuel ratio.** Raising AFR from 1.8 to 2.6 raises the throat temperature from 1385 to 1786 K. Char conversion rises from 11 to 99 % and efficiency from 46.6 to 69.8 %, while the gas LHV stays within 4.38–4.72 MJ/Nm^3. In this model efficiency keeps rising until the char is used up near AFR 2.6 (ER 0.42). Real gasifiers usually peak earlier because extra air burns product gas, which the throat sub-model represents only through equilibrium. This trend should be treated with caution beyond the tested range of AFR 1.86–2.37.
- **Feed rate.** Gas flow scales almost linearly with feed (28.9 to 58.2 Nm^3/h for 12–24 kg/h). Efficiency changes little (53.8–55.2 %), and the exit temperature rises from 669 to 938 K because a fixed wall loss becomes a smaller fraction of a larger throughput.

12 Limitations and uncertainty

1. **Throat temperature anchoring.** A mixed enthalpy balance gives about 1345 K at the reference condition. The model instead uses Jayah's ≈1604 K plus the balance-predicted *change*. The absolute throat temperature, and the size of the AFR, moisture and air-temperature effects, carry this assumption.
2. **Water-gas-shift rate.** The units of the literature expression are ambiguous. The fast interpretation would give a very different gas (Section 7.4).
3. **Calibration.** Three parameters are fitted: U to the 12.8 % loss, and η_B, η_{WG} to Jayah Table 5. The large effective U concentrates all measured loss in the hearth. The calibration run converts only 17.0 % of its char, below the 41–80 % implied by Jayah Table 4.
4. **Data gaps and inconsistencies in the paper.** Per-run feed rates and the basis of the measured gas flow are not given. Jayah Table 5 (2.5 cm) and Jayah Table 3 (Run 7, 5.5 cm) disagree on chip size. The chip size and moisture of the Jayah Fig. 5 trial are not stated. Jayah Table 4 implies an air–fuel ratio of about 3.0, outside Jayah Table 3's range.

5. **CH_4 and tar.** Methane is imposed rather than predicted. Tar is not modelled; it falls within Jayah's 7–9 % "unaccounted" material.
6. **2D model.** It resolves radial heat and species dispersion only. Velocity profiles, wall channelling, nozzle jets and bed bridging are not represented, and it is not CFD.
7. **Run-to-run variation.** The model does not reproduce the measured variation between runs ($R^2 \leq 0$).

13 Conclusions

1. The model was built from the paper's feedstock data (Jayah Tables 1–2), geometry and run conditions. It reproduces the calibration gas to 0.43 %-points (Jayah's model: 0.46).
2. Against the nine independent runs, the mean errors are 1.42 (CO), 2.01 (H_2) and 0.62 (CO_2) %-points, and 0.30 MJ/Nm^3 in LHV. H_2 is biased low and the run-to-run differences are not captured.
3. The axial gas temperature matches the six Jayah Fig. 5 measurements to 93 K, better than Jayah's own model (121 K). The gas and char equilibrate within about 7.6 cm of the throat.
4. The 2D model shows a hot core and a cold wall layer (1292 against 558 K at the exit). Core char is exhausted by about 11 cm while near-wall char is barely converted, so CO and H_2 are richest in the core.
5. The parametric study agrees with Jayah et al. on the direction of every effect and on heat loss being decisive. It disagrees on the optimum length: with 12.8 % loss, conversion stops after about 11–12 cm, so insulating the hearth, not lengthening it, is the recommended improvement.

Future work: measure per-run feed rate and gas flow on a stated basis; resolve the WGS kinetics and throat temperature with a non-premixed flame sub-model; add tar kinetics and a velocity/channelling model to the 2D formulation.

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A Model parameters

Table 9: Parameters as used in `src/config.py`.

Symbol	Description	Value	Status
U	wall heat-loss coefficient	114.4 $\mathrm{W\,m^{-2}\,K^{-1}}$	calibrated (12.8 % loss)
η_{B}	Boudouard effectiveness factor	0.474	calibrated (Jayah Table 5)
η_{WG}	water-gas effectiveness factor	0.142	calibrated (Jayah Table 5)
A_1, E_1	Boudouard intrinsic constants	3.616×10^8, 240 kJ/mol	as implemented
A_2, E_2	water-gas intrinsic constants	1.517×10^9, 241 kJ/mol	as implemented
$A_3\eta_3$, E_3	methanation	$4.189 \times 10^3 \times 0.05$, 130 kJ/mol	as implemented
k_4	water-gas shift	$2.78 \times 10^{-3}\, e^{-1510/T}$ $\mathrm{m^3/(mol\,s)}$	assumption (units)
T_{anchor}	throat temperature at AFR 2.20	1604 K	Jayah Fig. 5 (digitised)
ε	bed voidage	0.48	assumption
ρ_{char}	char particle density	250 $\mathrm{kg/m^3}$	assumption
CH_4 at throat	dry mole fraction	1.2 %	imposed (as in Jayah)
Pe_m, Pe_h	radial Péclet numbers (2D)	10, 8	packed-bed values

Because the rates are calibrated through η_j, the individual intrinsic pre-exponential factors are not claimed as independent literature values; only the products $\eta_j A_j$ are constrained by the data.

B Reproducing the results

```
pip install -r requirements.txt
python run_simulation.py              # all figures and tables (~30 s)
python -m pytest tests                # 20 tests
python -m src.calibration             # re-derive U, eta_B, eta_WG
python scripts/make_report_tables.py  # regenerate report numbers
```

Every simulated number in this report is generated from the files in `outputs/tables` by the script `make_report_tables.py` in `scripts`; none are typed by hand.